# Numerical Hessian Rank and Keplerian Eigenvalue Structure as Diagnostics for Periodic Orbits in the Equal-Mass Three-Body Problem

Andrew H. Bond[1, 2, *]

[1] *Department of Computer Engineering, San Jose State University, San Jose, CA 95192, USA*
[2] *Senior Member, IEEE; ORCID: 0009-0003-2599-6158*

(Dated: March 28, 2026)

We compute the Hessian of the gravitational three-body Hamiltonian in Jacobi coordinates across 3.24 million configurations and analyze its numerical rank, eigenvalue spectrum, and relationship to orbital periodicity. At configurations where the numerical rank of the potential-energy block $V_{qq}$ drops to 3 (using a singular-value threshold $\epsilon = 10^{-6}$), the three nonzero eigenvalues exhibit an exact 2:1:1 magnitude ratio—the curvature signature of a dominant Keplerian binary with a decoupled third body. This ratio is proven analytically from the Hessian of the $1/r$ potential and confirmed computationally to $< 10^{-12}$ relative error. We integrate 100,000 orbits on GPU and find that configurations at effective rank 9 (potential rank 3) show a periodicity rate of 5.5%, the highest among all rank levels. A gradient-boosted classifier trained on Hessian-derived features achieves cross-validated $F_1 = 0.86$ for periodicity prediction on a 51,000-configuration dataset, and an A*-discovered closed-form formula $\sqrt{n_{\text{clusters}}^2 + \lambda_{\min}^2} > \tau$ achieves $F_1 = 0.48$. The rank classification is threshold-dependent (vanishing at $\epsilon = 10^{-10}$) and the Keplerian structure is asymptotic (exact only in the decoupled limit), which we characterize through threshold sensitivity sweeps. We compare against separation ratio as a baseline and find that Hessian features provide complementary but not superior predictive power for periodicity.

## I. INTRODUCTION

The gravitational three-body problem has no general closed-form solution [1], but specific configurations—hierarchical triples, Lagrange points, choreographies—are approximately or exactly tractable. Identifying which configurations are tractable typically requires either perturbation theory [2], KAM analysis [3], or brute-force numerical integration [4]. Each approach is computationally expensive or requires analytical insight about the specific configuration.

We explore whether the *Hessian* of the Hamiltonian—computed once per configuration at negligible cost compared to orbit integration—contains sufficient information to predict orbital regularity. The Hessian encodes the local curvature of the energy landscape: its rank measures how many degrees of freedom are dynamically coupled, and its eigenvalue structure reflects the characteristic frequencies of the linearized motion.

Our approach is computational rather than theoretical. We make no claim that Hessian rank is a superior diagnostic to existing methods; rather, we characterize what it reveals and where it fails, as a contribution to the toolkit of three-body diagnostics.

* andrew.bond@sjsu.edu

## II. METHODS

### A. Jacobi Coordinates and Hessian Structure

We work in the center-of-mass-reduced 12-dimensional phase space $z = (\boldsymbol{\rho}_1, \boldsymbol{\rho}_2, \boldsymbol{\pi}_1, \boldsymbol{\pi}_2)$. The Hamiltonian separates as $H = T(\boldsymbol{\pi}) + V(\boldsymbol{\rho})$ with $T = |\boldsymbol{\pi}_1|^2/(2\mu_1) + |\boldsymbol{\pi}_2|^2/(2\mu_2)$.

The Hessian $C_{ab} = \partial^2 H/\partial z_a \partial z_b$ has exact block-diagonal structure:

$$C = \begin{pmatrix} V_{qq} & 0 \\ 0 & T_{pp} \end{pmatrix}, \tag{1}$$

where the position-momentum cross block is identically zero (since $H$ is separable in $q$ and $p$), and $T_{pp} = \text{diag}(\mu_1^{-1} I_3, \mu_2^{-1} I_3)$ is configuration-independent with eigenvalues $\{1/\mu_1, 1/\mu_1, 1/\mu_1, 1/\mu_2, 1/\mu_2, 1/\mu_2\}$. For equal masses, these are $\{2, 2, 2, 1.5, 1.5, 1.5\}$.

All variation in the Hessian structure comes from the $6 \times 6$ potential block $V_{qq}$, which we compute analytically (see Appendix). The total Hessian rank satisfies $\text{rank}(C) = \text{rank}(V_{qq}) + 6$.

### B. Numerical Rank

We define the *effective rank* at threshold $\epsilon$ as:

$$r_\epsilon(A) = |\{k : \sigma_k/\sigma_1 > \epsilon\}| \tag{2}$$

where $\sigma_k$ are the singular values of $A$ in descending order. Unless stated otherwise, we use $\epsilon = 10^{-6}$.

This is a numerical, not algebraic, quantity. At finite body separations, $V_{qq}$ is generically full rank (all six

eigenvalues nonzero). The "rank 3" designation means that three singular values are below $10^{-6}$ times the largest—not that they are exactly zero. Table I shows the sensitivity to $\epsilon$.

TABLE I. Fraction of configurations with effective rank < 12 at various thresholds ($N$ = 10,000 random sample, equal masses).

| Threshold $\epsilon$ | Fraction rank < 12 |
|---|---|
| $10^{-4}$ | 44.8% |
| $10^{-5}$ | 25.8% |
| $10^{-6}$ (default) | 8.2% |
| $10^{-7}$ | 2.9% |
| $10^{-8}$ | 1.0% |
| $10^{-10}$ | 0.0% |

### C. Configuration Sampling

We sampled $50\times50\times36\times36$ = 3,240,000 configurations for equal masses ($m_1 = m_2 = m_3 = 1$) on a grid in $(r_1, r_2, \theta, \phi)$ with logarithmic radial spacing $r \in [0.1, 100]$, where $r_1$ is the inner pair separation and $r_2$ is the distance from the third body to the inner pair's center of mass. Initial momenta are set for circular orbits.

### D. Orbit Integration

Orbits are integrated using a symplectic leapfrog (Störmer-Verlet) scheme with $\Delta t = 0.005$ for $8 \times 10^4$ steps ($T = 400$ natural time units). An orbit is classified as *periodic* if it returns to within 5% of its initial phase-space extent at any time after the first 10% of the integration. Energy conservation error is monitored; mean relative error is $\sim 10^{-3}$.

Integration is GPU-accelerated using CuPy on an NVIDIA Quadro GV100 (32 GB), processing up to 400,000 orbits simultaneously. The maximum batch size is determined automatically using the batch-probe library [5].

## III. RESULTS

### A. Rank Landscape

Table II shows the complete rank distribution.

### B. Keplerian Eigenvalue Structure

At configurations where $V_{qq}$ has effective rank 3, the three dominant eigenvalues exhibit a 2:1:1 magnitude ratio:

TABLE II. Hessian rank distribution (equal masses, $\epsilon = 10^{-6}$, $N$ = 3,240,000).

| Rank | Count | Fraction | $V_{qq}$ eff. rank |
|---|---|---|---|
| 6 | 24 | 0.001% | 0 |
| 7 | 1,316 | 0.04% | 1 |
| 8 | 716 | 0.02% | 2 |
| 9 | 156,756 | 4.84% | 3 |
| 10 | 74,124 | 2.29% | 4 |
| 11 | 32,972 | 1.02% | 5 |
| 12 | 2,974,092 | 91.79% | 6 |

$$|\lambda_1|/|\lambda_2| = |\lambda_1|/|\lambda_3| = 2.000\,000 \tag{3}$$

to better than $10^{-6}$ relative error across all 156,756 such configurations. The eigenvalue signs are $\{-\lambda, +\lambda/2, +\lambda/2\}$ where $\lambda \approx 2m_1m_2/r_1^3$.

*Proof.* In the limit where the third body is decoupled, $V \approx -m_1m_2/|\boldsymbol{\rho}_1|$. The Hessian of $-m/r$ in 3D evaluated at $\boldsymbol{\rho}_1 = (r_1, 0, 0)$ is:

$$\frac{\partial^2}{\partial x_i \partial x_j}\left(-\frac{m}{r}\right) = \frac{m}{r^3}\left(\frac{3x_ix_j}{r^2} - \delta_{ij}\right)$$

with eigenvalues $+2m/r^3$ (radial, multiplicity 1) and $-m/r^3$ (tangential, multiplicity 2). The magnitude ratio is exactly 2:1. At finite third-body distance, corrections of order $(r_1/r_2)^3$ perturb the eigenvalues but preserve the ratio to high accuracy when $r_2 \gg r_1$. □

This result is the Keplerian curvature structure expressed in the language of the coupling tensor. It is not new physics—it follows directly from the $1/r$ potential—but it provides a computationally cheap diagnostic: configurations exhibiting the 2:1 ratio are near-Keplerian hierarchical triples.

### C. Periodicity by Rank

Table III shows periodicity rates from the 100,000-orbit GPU survey.

TABLE III. Periodicity rate by effective Hessian rank (100,000 orbits, equal masses, circular initial momenta).

| Rank | Periodic | Total | Rate |
|---|---|---|---|
| 7–8 | 0 | 66 | 0.0% |
| 9 | 270 | 4,905 | 5.5% |
| 10 | 39 | 2,345 | 1.7% |
| 11 | 0 | 1,048 | 0.0% |
| 12 | 2,975 | 91,635 | 3.2% |

Rank 9 has the highest periodicity rate (5.5%), but rank 12 (generic, full rank) also shows 3.2% periodicity. The rank diagnostic enriches the periodic population by a factor of $5.5/3.2 \approx 1.7\times$ relative to the full-rank baseline—a modest but nonzero signal.

### D. Comparison to Separation Ratio Baseline

The simplest diagnostic for hierarchical stability is the separation ratio $r_2/r_1$. On the 51,000-configuration labeled dataset (three mass ratios), we compare Hessian-based predictors to this baseline:

TABLE IV. Periodicity prediction: Hessian features vs. baseline.

| Predictor | F1 |
|---|---|
| Random (base rate 5–10%) | ∼0.07 |
| $r_2/r_1 > 10$ | 0.29 |
| Frequency clusters $\geq 2$ | 0.29 |
| A*-discovered formula (Eq. 4) | 0.48 |
| Decision tree (depth 5, 9 features) | 0.58 |
| Gradient boosting (100 trees) | 0.86 |

The A*-discovered formula:

$$\sqrt{n_{\text{clusters}}^2 + \lambda_{\text{min}}^2} > \tau \tag{4}$$

achieves F1 = 0.48, outperforming the separation ratio baseline (F1 = 0.29) but well below the gradient-boosted ensemble (F1 = 0.86). The frequency cluster count alone matches the separation ratio—it is a Hessian-derived equivalent of the same information, not a new signal.

### E. Machine Learning Predictor

The gradient-boosted classifier (F1 = 0.86, 5-fold cross-validated) uses 9 Hessian-derived features. Feature importances reveal that *energy* (38%), *log separation ratio* (18%), and *log frequency ratio* (12%) dominate. Tensor rank contributes only 10%. The decision tree identifies the primary split as rank $\leq 9.5$ (importance 46%), recovering the rank threshold automatically.

The gap between the closed-form formula (F1 = 0.48) and the ensemble (F1 = 0.86) quantifies the irreducible complexity of the periodicity prediction task: 38% of the predictive structure requires nonlinear feature interactions that no depth-4 formula can express.

## IV. DISCUSSION

The Hessian rank landscape provides a computationally cheap (97× faster than integration) map of the three-body configuration space that correlates with but does not strongly predict orbital periodicity. The Keplerian 2:1 eigenvalue ratio is a clean analytical result but amounts to detecting near-Keplerian hierarchical structure—information also available from the separation ratio.

The principal value of the Hessian approach is in the machine learning setting: the 9 Hessian-derived features, when combined nonlinearly, achieve F1 = 0.86 for periodicity prediction, substantially outperforming any single diagnostic. This suggests that the Hessian encodes complementary information about the energy landscape curvature that simple geometric diagnostics (like separation ratio) miss.

### A. Limitations

The rank classification is threshold-dependent (Table I) and should not be interpreted as exact algebraic rank. All "rank 9" configurations have six nonzero singular values; three are merely small. The 2:1 eigenvalue ratio is exact only in the fully decoupled limit and approximate (with $(r_1/r_2)^3$ corrections) at finite separations. The periodicity classification depends on integration parameters (timestep, duration, return threshold). Results are shown for equal masses only; generalization to unequal masses shows qualitatively similar patterns but quantitatively different rank distributions. The 4-body generalization (3,000 configurations, equal masses) found 0% periodicity, indicating that the approach does not trivially extend to $N > 3$.

## V. CONCLUSION

The Hessian of the three-body Hamiltonian provides a fast, analytically grounded diagnostic for near-Keplerian hierarchical structure via its numerical rank and 2:1 eigenvalue ratio. As a standalone periodicity predictor, its power is modest (F1 = 0.48 for the best formula, 1.7× enrichment at rank 9). Combined with energy and separation features in a machine learning ensemble, it contributes to a strong predictor (F1 = 0.86). The rank landscape atlas (3.24 million configurations) and 100,000-orbit GPU dataset are publicly available for further analysis.

## ACKNOWLEDGMENTS

Computations performed on a dual-GV100 workstation using CuPy for GPU-accelerated batch integration. Optimal GPU batch sizes determined using the batch-probe library [5]. Code and data: `https://github.com/ahb-sjsu/astar-symbolic-discovery`.

[1] H. Poincaré, *Les méthodes nouvelles de la mécanique céleste* (Gauthier-Villars, Paris, 1892).
[2] C. D. Murray and S. F. Dermott, *Solar System Dynamics* (Cambridge University Press, 1999).
[3] V. I. Arnold, Russ. Math. Surv. **18**, 85 (1963).
[4] M. Šuvakov and V. Dmitrašinović, Phys. Rev. Lett. **110**, 114301 (2013).
[5] A. H. Bond, batch-probe: Find the maximum batch size that fits in GPU memory, v0.3.0 (2026). `https://pypi.org/project/batch-probe/`